

Week 02

Ex 2

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①

$$x_1(n) = \{-3, 2, \underset{\uparrow}{2}, 1, 0, 4, -1\}$$

$$x_2(n) = \{-1, 2, -3, \underset{\uparrow}{-3}, 0\}$$

$$a) y_1(n) = x_1(n) + x_2(n) = \{-1, -1, -1, \underset{\uparrow}{-1}, 1, 0, 4, -1\}$$

$$b) y_2(n) = \frac{1}{3}x_1(n) - \frac{2}{3}x_2(n) = \left\{\frac{2}{3}, -1 - \frac{2}{3} \cdot 2, \frac{2}{3} + 2, \frac{2}{3} + 2, \frac{1}{3}, 0, \frac{4}{3}, -\frac{1}{3}\right\}$$

$$\Rightarrow y_2(n) = \left\{\frac{2}{3}, -\frac{7}{3}, \frac{8}{3}, \frac{8}{3}, \frac{1}{3}, 0, \frac{4}{3}, -\frac{1}{3}\right\}$$

$$c) y_3(n) = \{0, -6, -6, \underset{\uparrow}{-6}, 0, 0, 0, 0\}$$

②

Let's say we have an arbitrary signal  $x[n]$ . Then we can construct an even signal  $x_e[n]$ :

$$x_e[n] = \frac{1}{2} (x[n] + x[-n])$$

This signal is even because:

$$\begin{aligned} x_e[n] &= \frac{1}{2} (x[-n] + x[-(-n)]) = \frac{1}{2} (x[-n] + x[n]) \\ &= \frac{1}{2} (x[n] + x[-n]) = x_e[n] \quad \blacksquare \end{aligned}$$

Similarly for  $x_o[n]$ :

$$\begin{aligned} x_o[n] &= \frac{1}{2} (x[n] - x[-n]) = \frac{1}{2} (x[-n] - x[n]) \\ &= -\frac{1}{2} (x[n] - x[-n]) = -x_o[n] \end{aligned}$$

$$\Rightarrow x_o[-n] = -x_o[n] \quad \blacksquare$$

We further see:

$$\begin{aligned} x[n] &= x_e + x_o = \frac{1}{2} (x[n] + x[-n]) + \frac{1}{2} (x[n] - x[-n]) \\ &= \frac{1}{2} x[n] + \frac{1}{2} x[n] + \frac{1}{2} x[-n] - \frac{1}{2} x[-n] = \underline{\underline{x[n]}} \end{aligned}$$



③

$$a) y[n] = x[n] + y[n-1]$$

Linear:

$$y_1[n_1] + y_2[n_2] = x_1[n_1] + y_1[n_1-1] + x_2[n_2] + y_2[n_2-1]$$

$$\Rightarrow x_1[n_1] + x_2[n_2] + y_1[n_1-1] + y_2[n_2-1]$$

$$\Rightarrow (x_1[n_1] + x_2[n_2]) + (y_1[n_1-1] + y_2[n_2-1])$$

Linear  $c \cdot y[n] = c \cdot x[n] + c y[n-1]$

Causal

Not static  $\rightarrow$  Dynamic because  $y[n] \neq f(x[n])$

Time invariant:

$$y[n-k] = x[n-k] + y[n-k-1]$$

We have: Linear, Time-Invar, Causal, Dynamic

③

$$b) y[n] - y[n+1] = 12x[n+2]$$

$$\Rightarrow y[n] = 12x[n+2] + y[n+1]$$

Not static, so dynamic

Linear:

$$y_1[n] = 12x_1[n+2] + y_1[n+1]$$

$$\Rightarrow y_2[n] = 12x_2[n+2] + y_2[n+1]$$

$$\Rightarrow y_3[n] = 12(x_1[n+2] + x_2[n+2]) + y_3[n+1]$$

$$\Rightarrow y_3[n] = y_1 + y_2 \quad \square \quad \underline{\underline{\text{Linear}}}$$

• Time-invariant

• Not causal, since  $y[n] = f(x[n+2])$

So we have: Linear, Time-inv, Not causal, dynamic



④

$$y[n] - 3y[n-1] + 6y[n-2] = x[n-1]$$

$$\delta[n] = \{1\} \text{ with } \delta[n \neq 0] = 0$$

$$\Rightarrow y[0] - 3y[-1] + 6y[-2] = x[-1] \Rightarrow \underline{y[0] = 0}$$

$$\Rightarrow y[1] - 3y[0] + 6y[-1] = x[0] \Rightarrow \underline{y[1] = 1}$$

$$\Rightarrow y[2] - 3y[1] + 6y[0] = x[1] \Rightarrow \underline{y[2] - 3 = 0}$$

$$\Rightarrow \underline{y[2] = 3}$$

$$\Rightarrow y[3] - 3y[2] + 6y[1] = x[2] \Rightarrow y[3] - 9 + 6 = 0$$

$$\Rightarrow \underline{y[3] = 3}$$

$$\Rightarrow y[4] - 3y[3] + 6y[2] = x[3] \Rightarrow y[4] - 3 \cdot 3 + 6 \cdot 3 = 0$$

$$\Rightarrow \underline{y[4] = -9}$$

$$\Rightarrow \underline{\underline{h[n] = \{0, 1, 3, 3, -9\}}}$$